SCANFISH

CABLE CONFIGURATION

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1 Abstract

Hydrodynamic analysis of the tow cable has an important role when configuring a ScanFish system. The cable catenary including cable properties, cable depth corresponding to a given cable length and cable tension are parameters that affects the ScanFish system. This document assumes a Ø11.43 mm cable, which is the preferred cable size when towing a ScanFish.

The cable is analysed at different pay-outs and speeds when towing the ScanFish and presents the catenary profile that compares the depth and position after the vessel.

2 Towing characteristics

This section includes graphical figures that represent the towing characteristics of the ScanFish for various cable lengths, speeds and depths.

The following ScanFish layback figures are provided to assist in towing the system. Please note the following points if the figures are used as reference:

- The standard ScanFish currently supports depths to 400 meters.
- The results listed in these tables are calculated using the EIVA NaviCat cable program with the best-estimated parameters for vehicle weight, drag, lift and buoyancy, as well as cable weight and stiffness. EIVA disclaims any liability for consequential damage resulting from using the data in the tables.
- The values contained in these figures are subject to change and have been assigned an error of +/-10%. Keep this in mind, especially if within 10% of the limits of any parameter.
- These guidelines are to be used with caution until verified by actual sea trials.

The figures contained within this document have been generated under the following assumed conditions:

- Sea state = 0 (no vessel heave motion)
- Solutions are for steady-state tow vessel speeds (4 kts to 10 kts, 2-kt increments)
- The cable used is Ø11.43
- Cable length 50 m to 3,600 m

Ocean current will be a factor for virtually all cases in which the system is used. By following the simple procedure below, the existing figures can be used for predicting the ScanFish depth when ocean currents are present (assuming the ocean current is uniform with depth).

The speed specifications of the system are measured in speed through water (STW). Therefore, if you are going with the current, you need to increase the vessel speed by the same amount as the current.

For example, for the speed through water of the ScanFish to be 3 knots, when you are sailing with the current at 2 knots, then you need to have a vessel speed of 5 knots speed over ground (SOG).

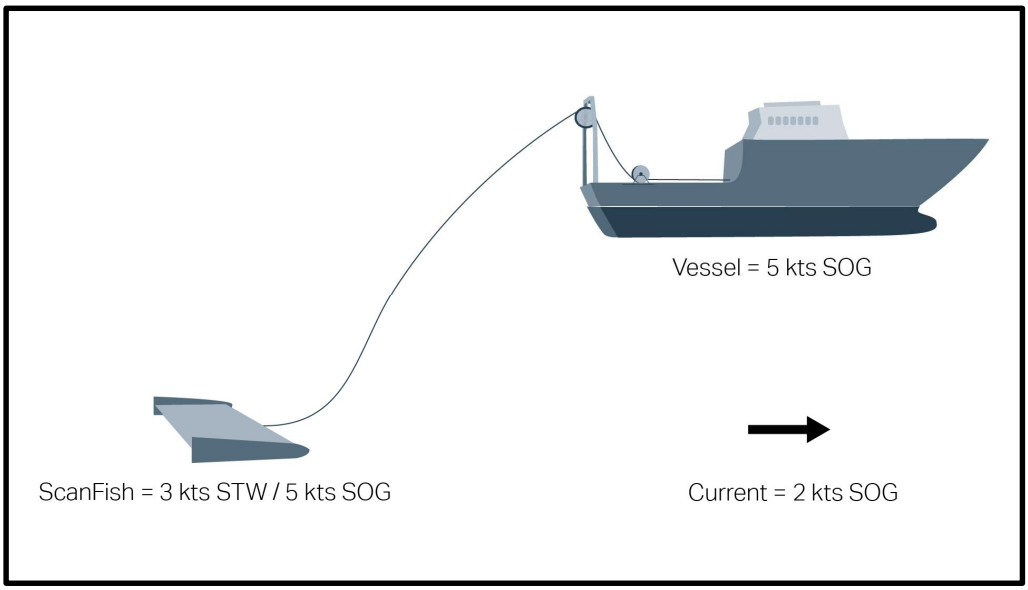

Figure 1 Vessel speed when sailing with the current to achieve a target STW for the ScanFish

And the other way around, as STW speed changes, drag will change accordingly.

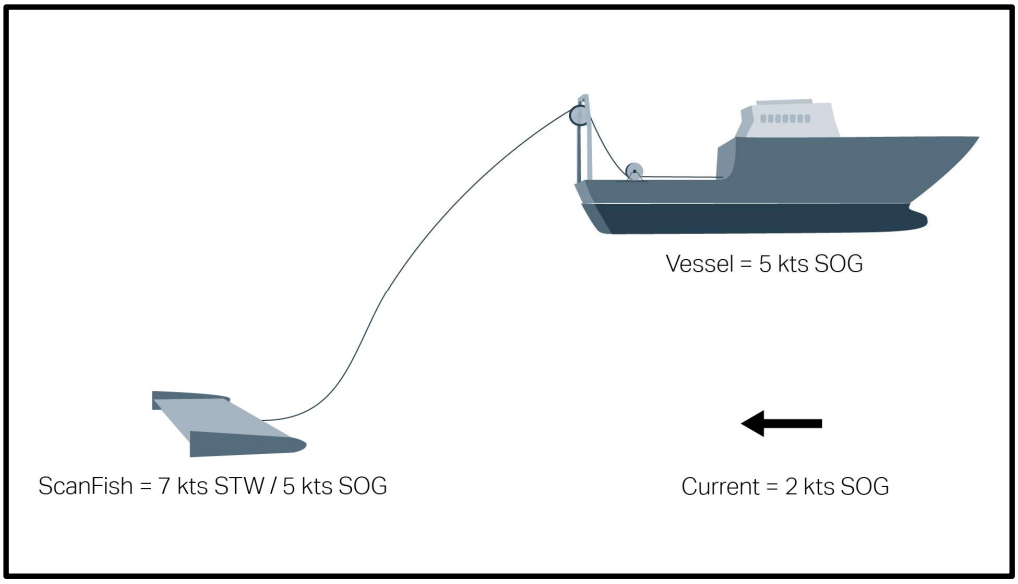

Figure 2 Vessel speed when sailing against the current to achieve a target STW for the ScanFish

2.1 Fairing

A fairing is a structure attached to a towed cable designed to streamline the flow around the cable. Cables are faired primarily for two reasons:

- To reduce normal drag and thus achieve more depth for a given cable scope and speed.
- To eliminate cable vibration caused by vortex shedding, commonly known as cable strum.

Hair fairing is a hair like flexible structure attached to cable that assumes a streamlined shape when towed. Drag coefficient Cd: 0.6.

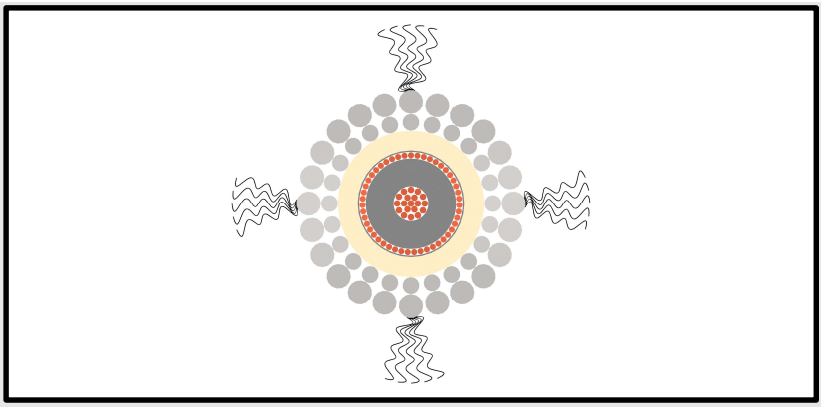

Figure 3 A hair-faired cable

Ribbon fairing is a cable with flexible ribbons attached to it that assumes a streamlined shape when towed. Drag coefficient Cd: 0.2.

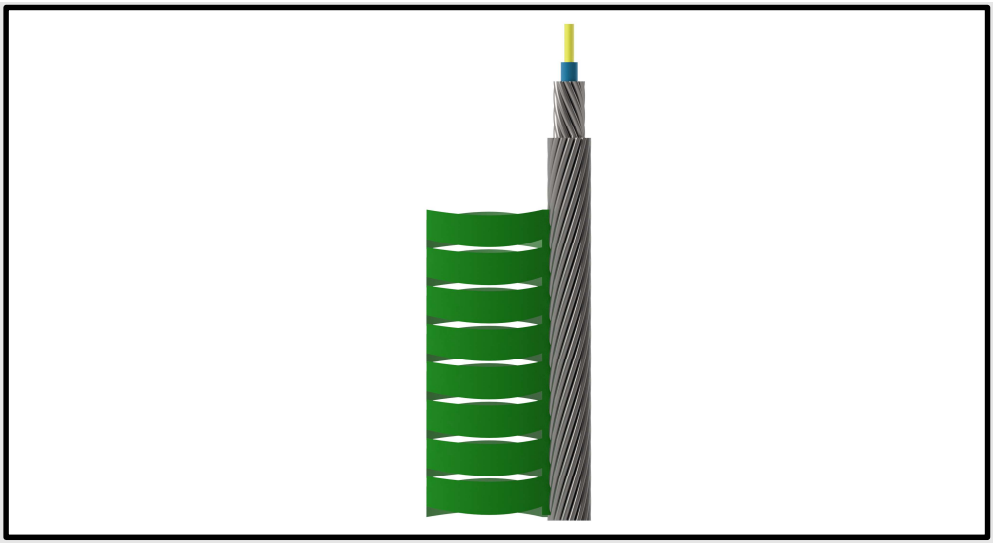

Figure 4 A ribbon-faired cable

2.2 Layback and cable length – tow cable with no fairing

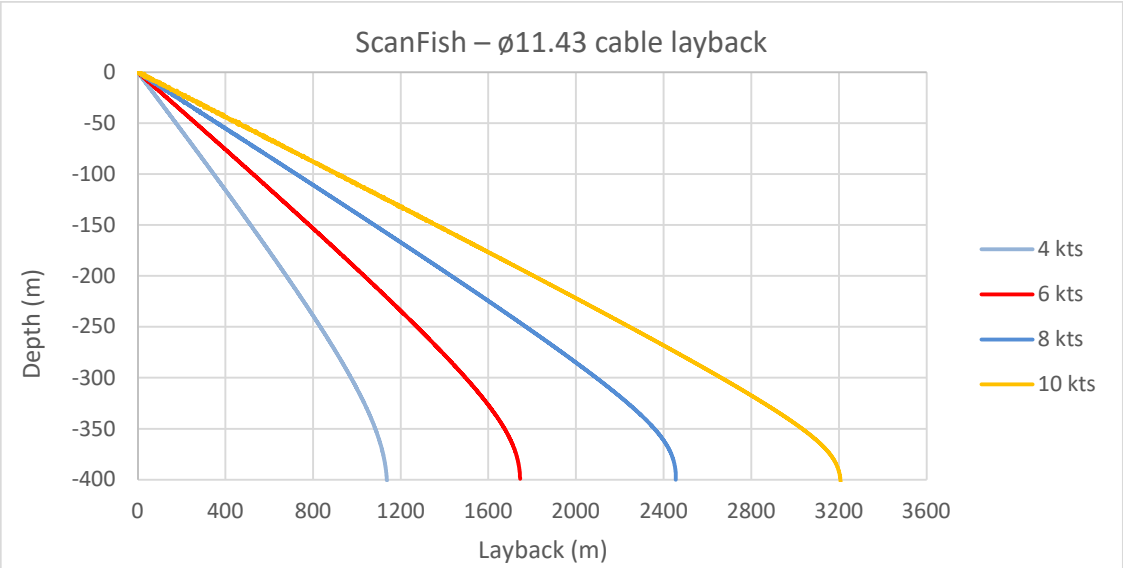

Figure 5 ScanFish tow cable layback versus depth (cable with no fairing)

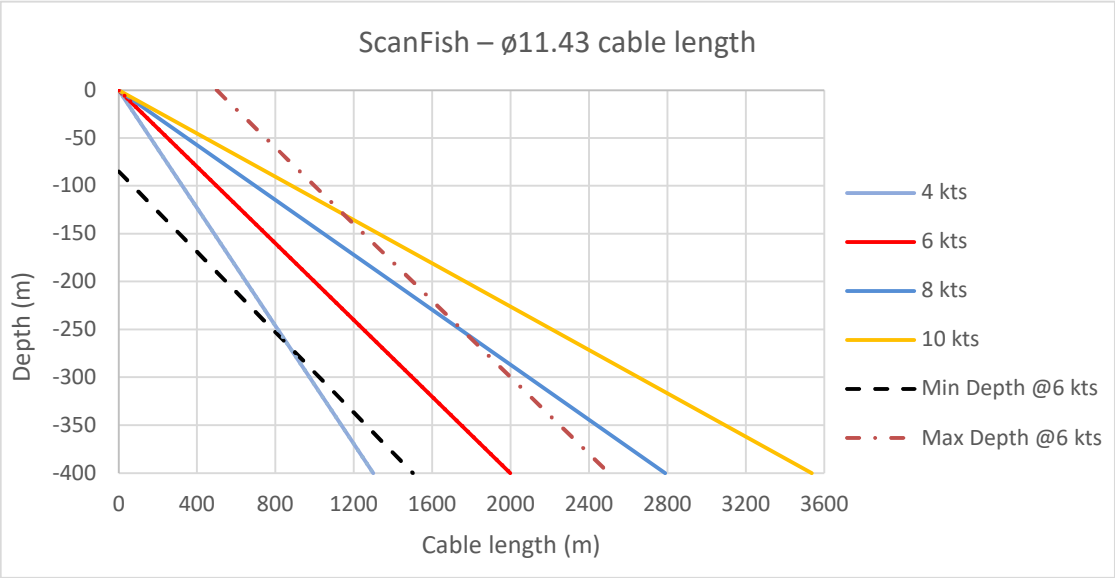

Figure 6 ScanFish tow cable length versus depth (cable with no fairing)

2.3 Layback and cable length – tow cable with 200 m hair fairing

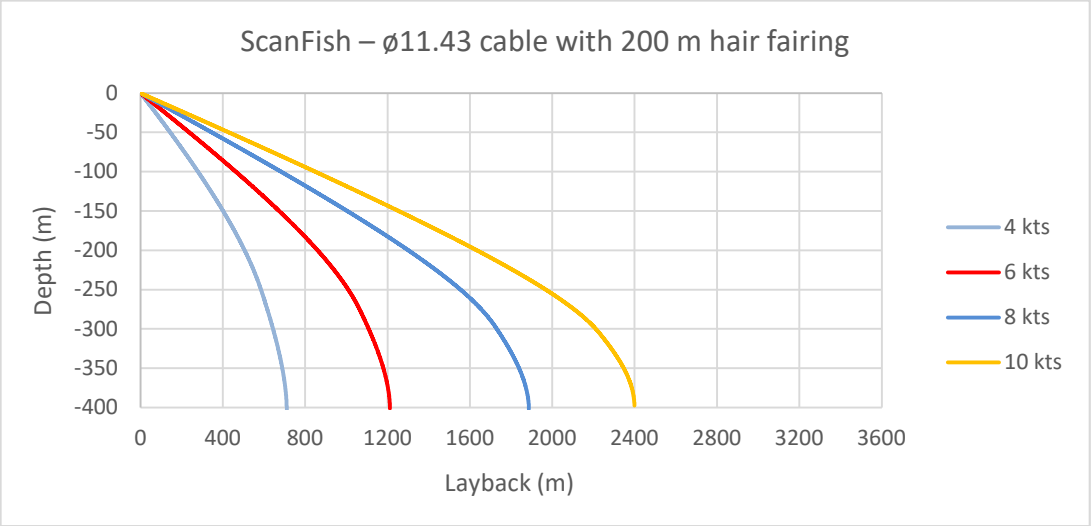

Figure 7 ScanFish tow cable layback versus depth (200 m hair-faired cable)

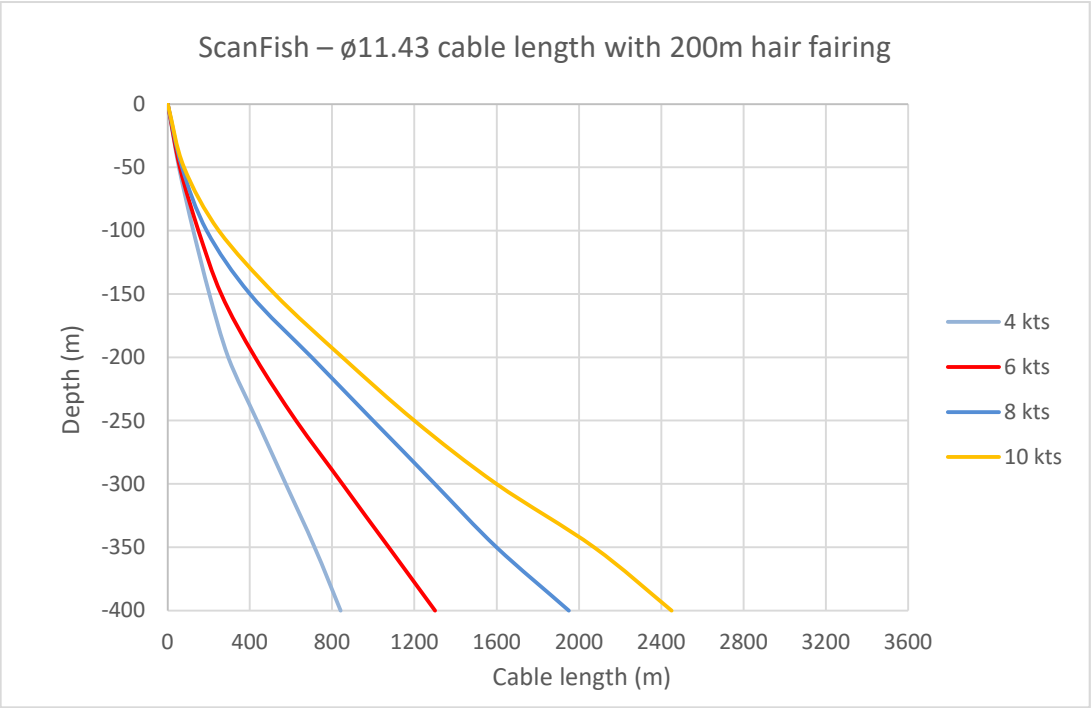

Figure 8 ScanFish tow cable length versus depth (200 m hair-faired cable)

2.4 Layback and cable length – tow cable with full hair fairing

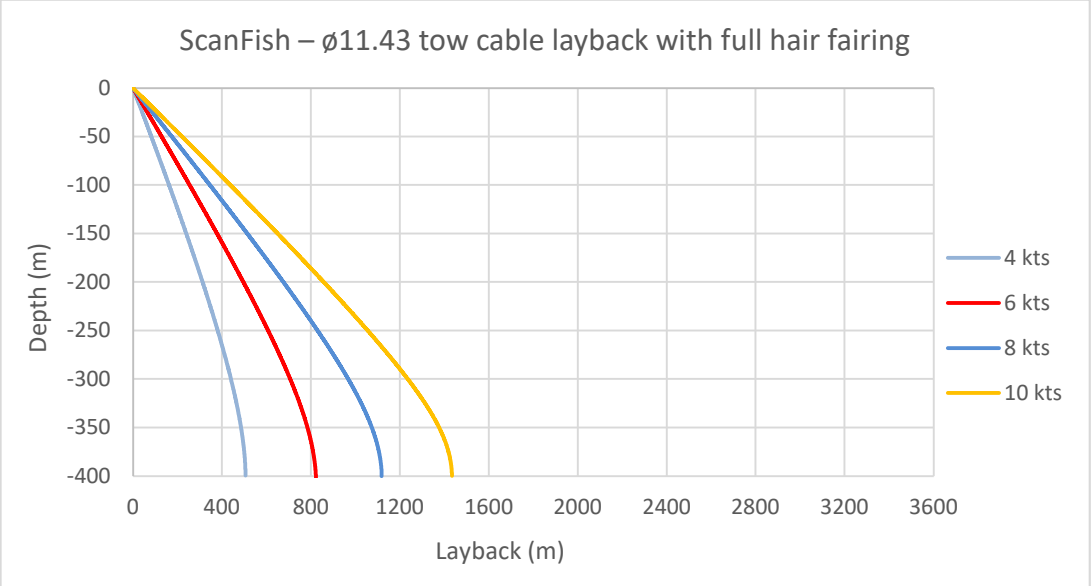

Figure 9 ScanFish layback versus depth (fully hair-faired cable)

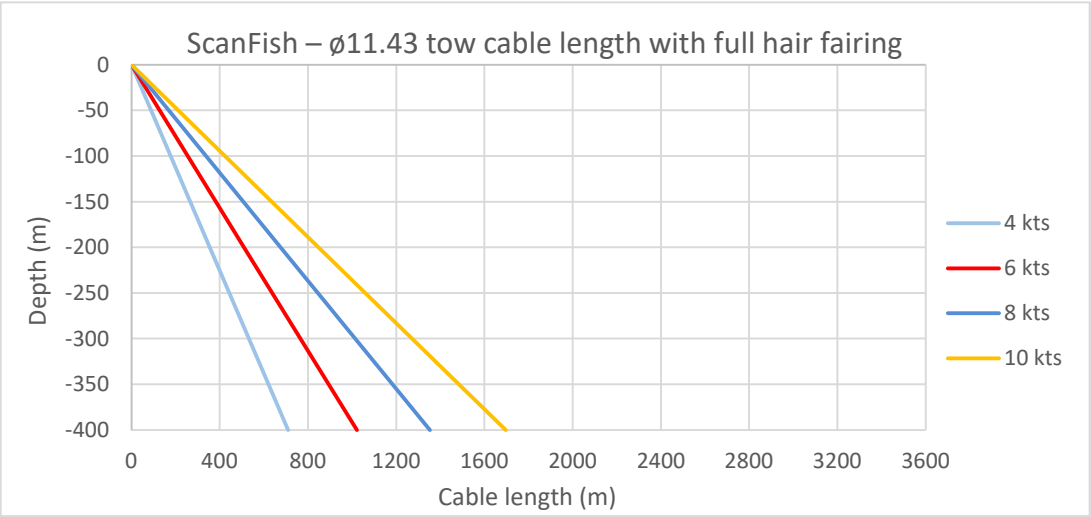

Figure 10 ScanFish tow cable length versus depth (fully hair-faired cable)

2.5 Layback and cable length – tow cable with 200 m ribbon fairing

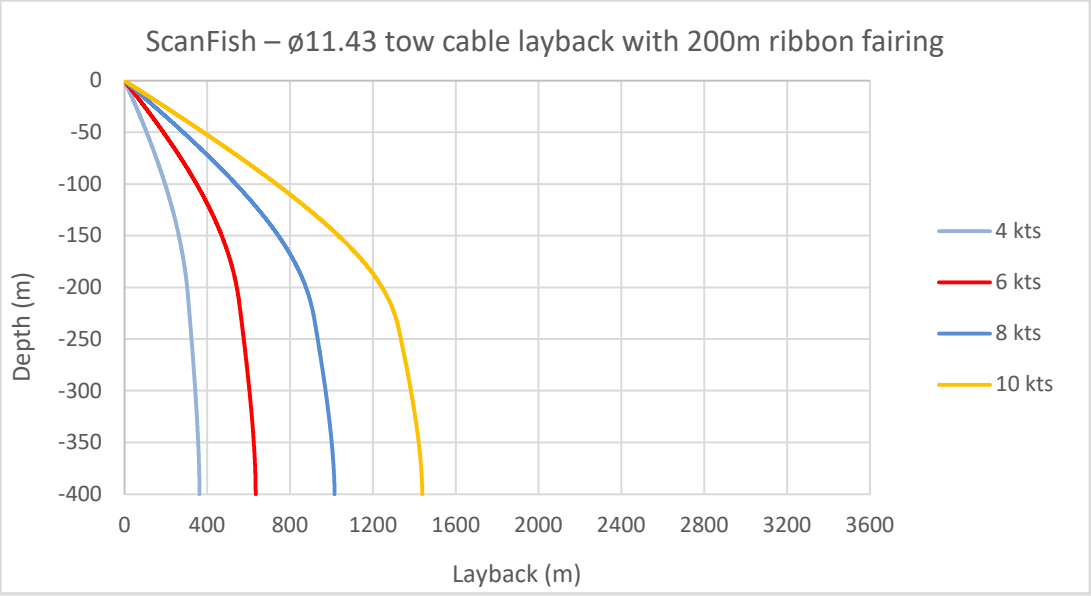

Figure 11 ScanFish layback versus depth (200 m ribbon-faired cable)

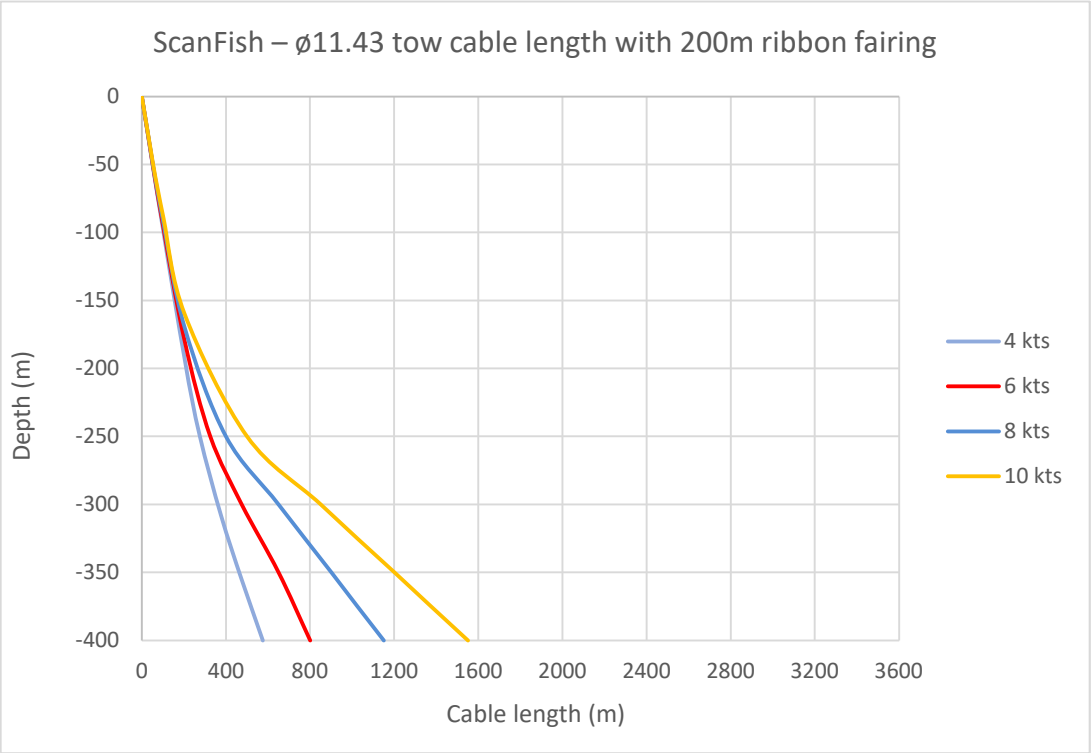

Figure 12 ScanFish tow cable length versus depth (200 m ribbon-faired cable)

2.6 Layback and cable length – tow cable with full ribbon fairing

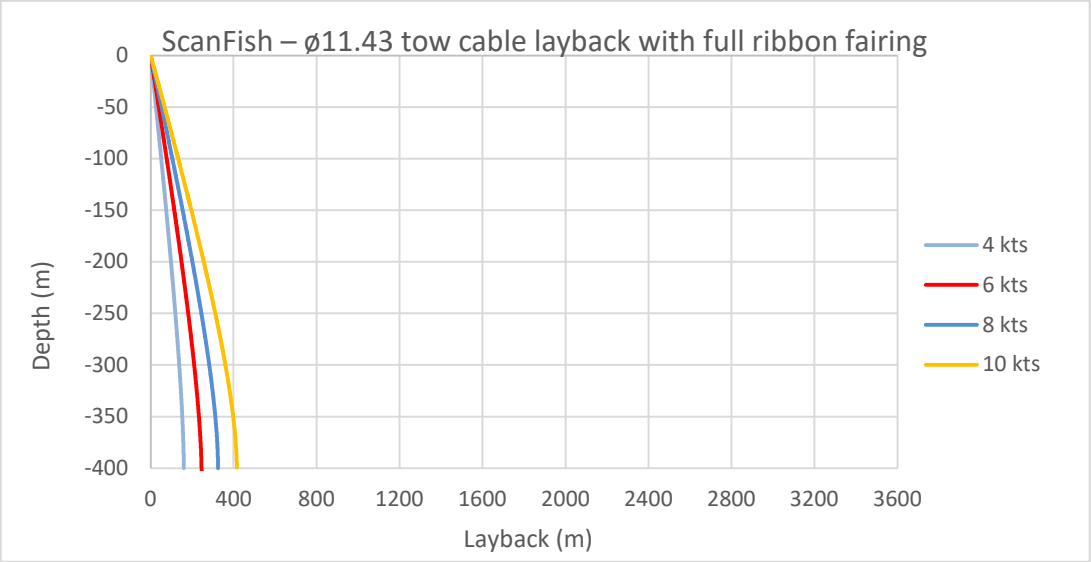

Figure 13 ScanFish layback versus depth (fully ribbon-faired cable)

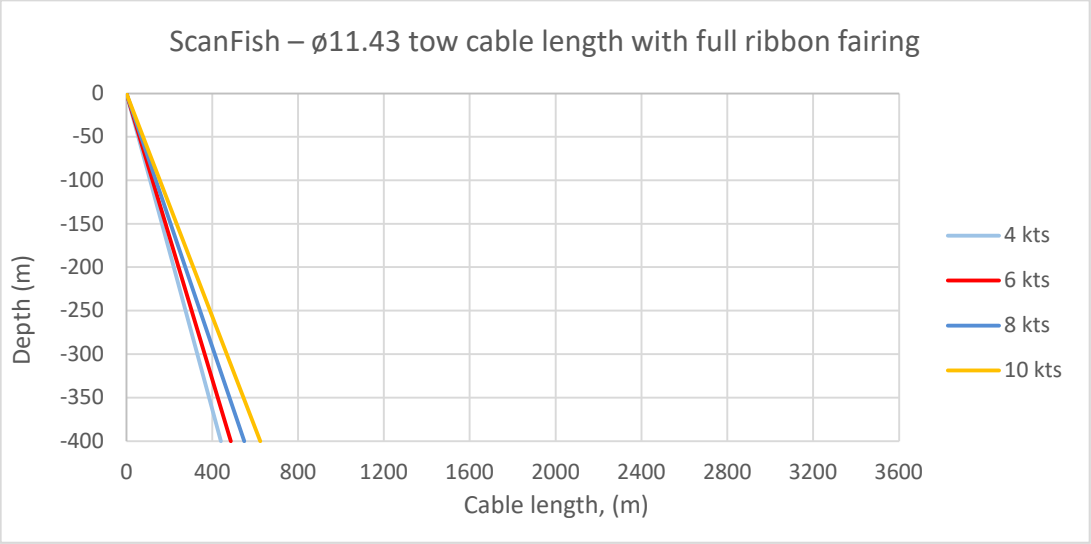

Figure 14 ScanFish tow cable length versus depth (fully ribbon-faired cable)

3 Cable tension with ScanFish

When the cable with a ScanFish is towed through the water, there is a tension in the cable and the winch.

This tension is a result of a combination of factors: the depth of the cable, the STW of the ScanFish, the drag on the cable, the swept area of the cable and the fairing used on the cable.

The following graphs show the cable tension by depths at different velocities for cables with and without fairing.

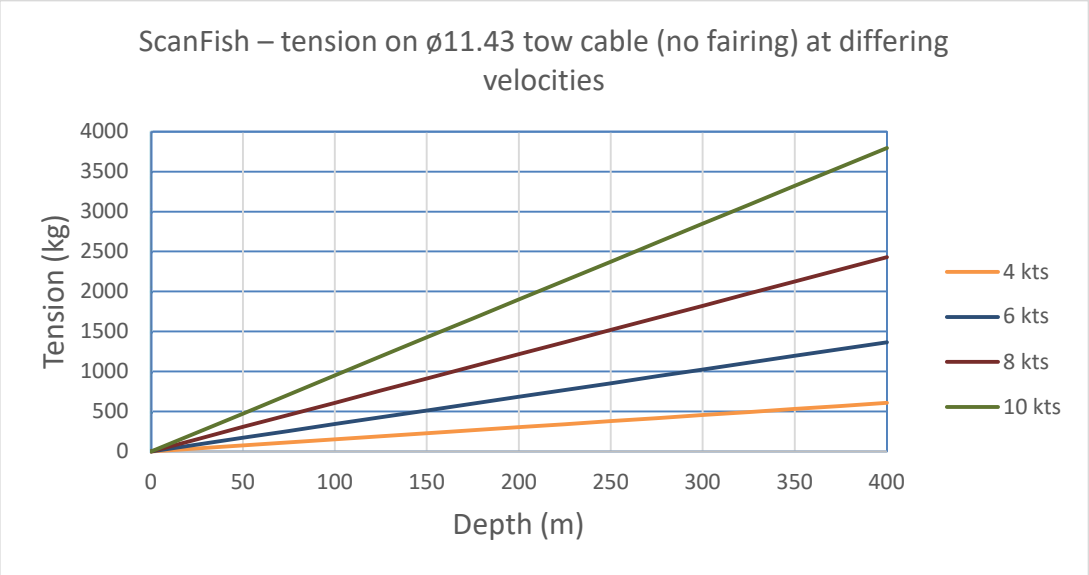

Figure 15 ScanFish tow cable tension v depth at different velocities (cable with no fairing)

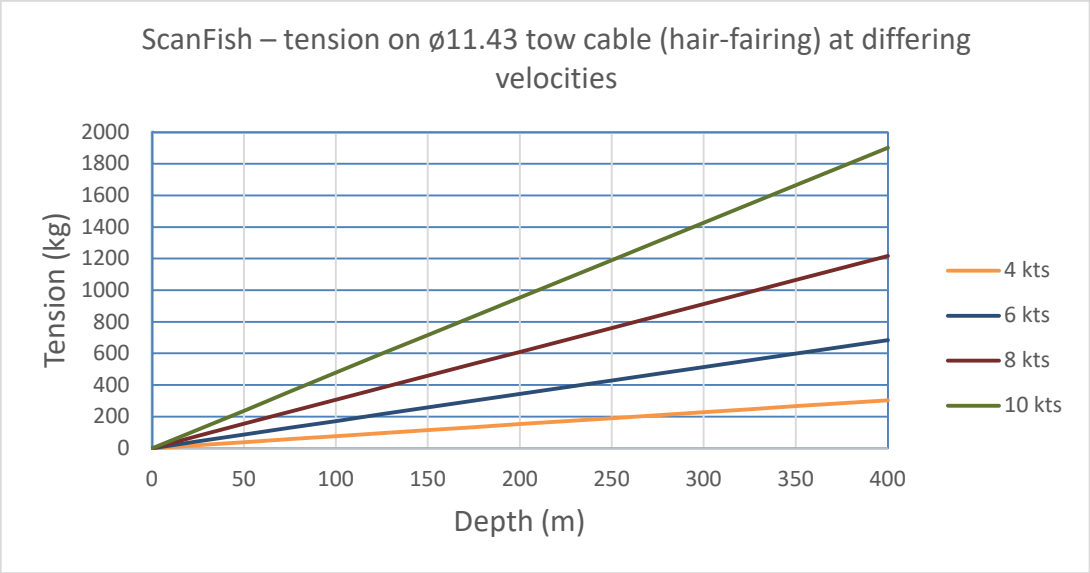

Figure 16 ScanFish cable tension v depth at different velocities (Ø11.43 hair-faired cable)

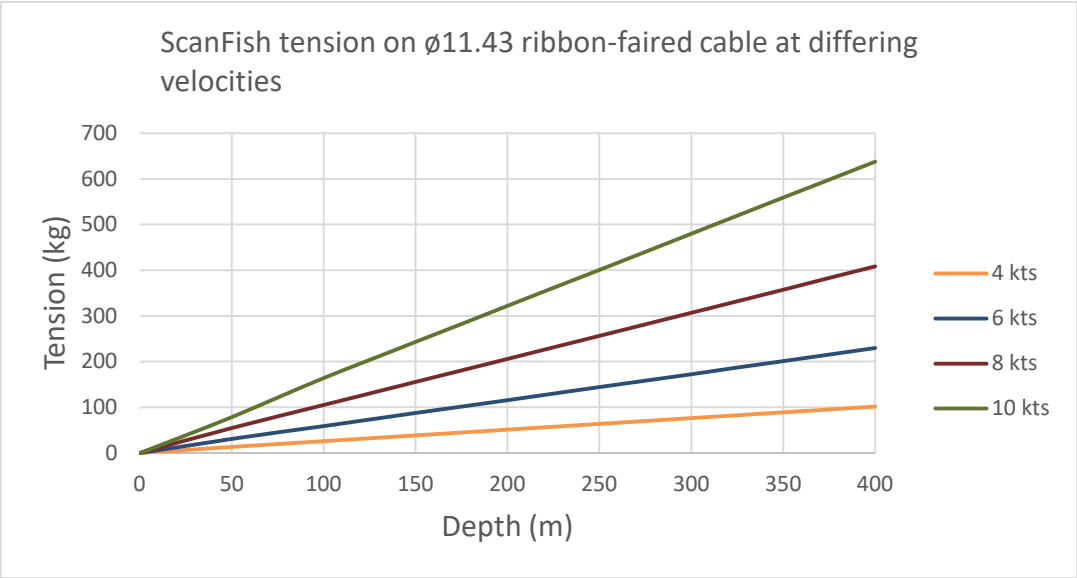

Figure 17 ScanFish cable tension v depth at different velocities (Ø11.43 ribbon-faired cable)

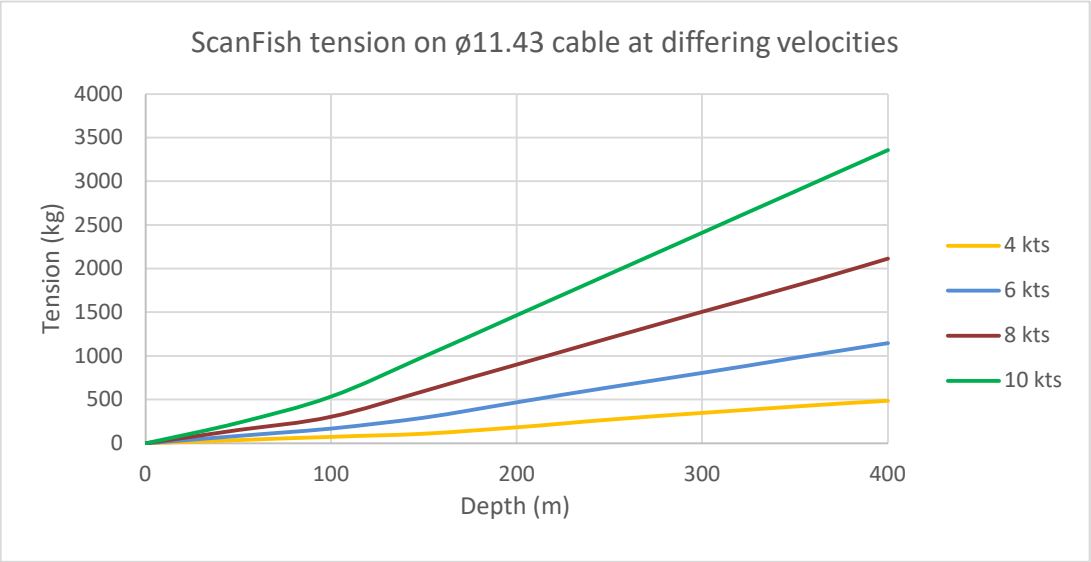

Figure 18 ScanFish tow cable tension v depth at different velocities (Ø11.43 hair-faired cable – first 200 m)

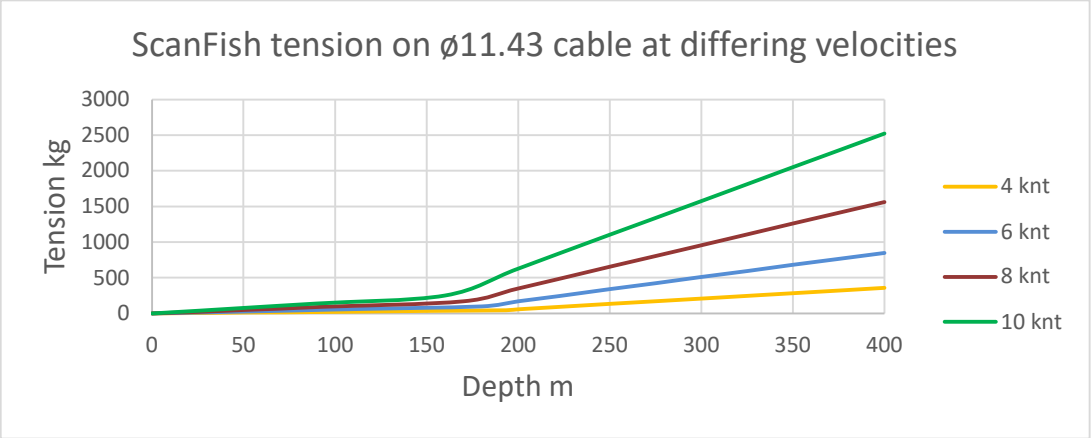

Figure 19 ScanFish tow cable tension v depth at different velocities (Ø11.43 ribbon-faired cable – first 200 m)

4 Depth versus undulation rate (m/s)

The ScanFish is designed to undulate between 1 and 400 m. The tables below show the undulation rate, for undulating in different undulating windows. The windows shown below are between 100 m and 150 m, at various depths. Other windows are available, eg 1-400 m. This of course gives another undulation rate. The tables are created as worst-case scenarios – this means the time has been taken from undulating using the winch at full speed, without ScanFish flap movement. The winch has a variable-speed drive which the operator must set to 100% to be able to achieve the presented undulations rates. By using the flaps, the undulation rate (m/s) will increase.

Window 100 m undulation rate (m/s) between 1 m to 100 m (Worst-case scenario):

Undulation rate (m/s) at differing speeds (kts) for a 100 m undulation window at depth between 1–100 m using an OE10.4LV winch

Fairing type	4 kts	6 kts	8 kts	10 kts
No fairing	0.34 m/s	0.22 m/s	0.15 m/s	0.12 m/s
Hair Fair	0.77 m/s	0.47 m/s	0.35 m/s	0.27 m/s
Ribbon Fair	1.01 m/s	0.93 m/s	0.82 m/s	0.73 m/s
200 m Fairing	1.13 m/s	1.13 m/s	1.13 m/s	1.13 m/s

Table 1 Undulation rate in m/s of 100 m rise and fall between depths of 1 to 100 m

Window 150 m undulation rate (m/s) between 100 m to 250 m (Worst-case scenario)

OE10.4LV winch	Speed			
	4 kts	6 kts	8 kts	10 kts
No fairing	0.32 m/s	0.19 m/s	0.13 m/s	0.10 m/s
Hair Fair	0.75 m/s	0.46 m/s	0.32 m/s	0.25 m/s
Ribbon Fair	1.00 m/s	0.91 m/s	0.80 m/s	0.71 m/s
200 m Fairing	1.00 m/s	0.81 m/s	0.63 m/s	0.47 m/s

Table 2 Undulation rate in m/s of Rise and Fall 150 m between depths of 100 to 250 m

Window 150 m undulation between 250 m to 400 m (Worst-case scenario)

OE10.4LV winch	Speed			
	4 kts	6 kts	8 kts	10 kts
No fairing	0.30 m/s	0.17 m/s	0.11 m/s	0.08 m/s
Hair Fair	0.73 m/s	0.43 m/s	0.30 m/s	0.22 m/s
Ribbon Fair	1.10 m/s	1.05 m/s	1.02 m/s	0.97 m/s
200 m Fairing	0.73 m/s	0.33 m/s	0.26 m/s	0.18 m/s

Table 3 Undulation rate in m/s of Rise and Fall 150 m between depths of 100 to 250 m

5 Recommended cables

The following tow cables are recommended when towing a ScanFish:

- Rochester Type A302799 Ø11.43 mm coax cable
- Rochester Type EQ10954 Ø11.33 mm fiber optic and coax cable
- Hydrocable systems Type CS8750 Ø11.8 mm fiber optic and 2+2 cores cable

Please contact EIVA if you would like more information on these cables.